StockGPT on NSE: Language-Model Return Prediction for Indian Equities

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Abstract

This paper tests whether StockGPT (Mai, 2024), a small GPT-style transformer that forecasts stock returns from past daily returns alone, works on Indian equities. Using a survivorship-free NSE dataset of 9.9 million daily records across 4,047 stocks from 1995 to 2025, the model is trained on early data and tested once on an untouched 2019–2025 period. A model trained on data up to 2010 and never updated earns a rank IC of 0.036 (t = 3.9) out of sample. Retrained on data up to 2018, a monthly long-only portfolio of its top 5% forecasts returns 41.1% a year after 57 bp round-trip costs, against 19.8% for the equal-weighted universe (excess t = 3.44, net Sharpe 1.15, maximum drawdown 34%), turning ₹1 lakh into ₹11.0 lakh against ₹2.9 lakh. The edge is concentrated in smaller stocks and in 2020–2021: among the roughly 180 large, liquid F&O stocks it disappears.

1. Introduction

Mai (2024) showed that a language model trained on sequences of US daily stock returns, treated like words in a sentence, can rank next-period returns well enough to build a profitable portfolio. If the result reflects a general feature of how prices move, it should carry over to other markets. India is a useful test: it has a large, fast-growing listed universe, a high share of small and thinly traded stocks, and a market structure quite different from the US.

This paper makes three contributions:

1. **A clean Indian dataset.** A survivorship-free NSE daily panel built from raw exchange files, with corporate-action adjustments and delisted stocks kept until their exit.
2. **A disciplined out-of-sample test.** The original paper's published settings, separate training, selection and test periods, and a single run on the final test period.
3. **An honest account of where the edge comes from.** Results by year, by tradable universe (F&O stocks only) and after realistic costs, so a practitioner can judge whether the signal can be traded.

2. Data

The dataset holds 9.9 million daily records for 4,047 NSE stocks from 1995 to 2025, including stocks that were later delisted or merged.

- **Source.** NSE daily bhavcopy files, downloaded in five-year blocks. Commercial vendors were rejected because they drop delisted stocks; Interactive Brokers, for example, has no history for DHFL, Satyam, Kingfisher, Jet Airways or HDFC Ltd and starts NSE coverage around 1999.
- **Adjustments.** Prices are adjusted for splits, bonuses and dividends, with manual checks on edge cases. Adjusted prices match Interactive Brokers for over 99% of observations where its data is clean.
- **Delistings.** A stock stays in the universe until its last trading day and exits at its last traded price.
- **Liquidity filter.** Each month, stocks are screened on 60-day median daily turnover, using past data

only. Historical market capitalisation is not available for most of the sample, so turnover serves as the size measure throughout.

- **Pump-and-dump filter.** A day is "locked" when the high equals the low with a move of 5% or more. A stock is excluded if more than half of its last 60 trading days were locked.

After filtering, the test period covers about 1,400 stocks a month.

3. Method

- The model follows Mai (2024) exactly: no hyperparameters were tuned on Indian data.
- **Model.** Daily returns are grouped into bins and treated as tokens, and a GPT-style transformer learns to predict the next token from the sequence before it. The base model has about 0.93 million parameters (embedding size 128, 4 blocks of 4 attention heads, dropout 0.2, batch 64, 10,000 training steps). An upgraded version with modern components (rotary position embeddings, RMSNorm, SwiGLU) was also tested. Each model was trained once, with one random seed, on GPU.
- **Periods.** Gaps at the period boundaries stop data from one period leaking into the next.

Period	Years	Use
Training	1995–2010	Fit the models
Selection	2011–2018	Compare models and benchmarks; no trading
Final test	2019–2025	One run of the chosen model, 83 months

Benchmarks. A model must beat two simple signals, short-term reversal and 12-1 momentum, plus two machine-learning alternatives: LightGBM on lagged returns and Amazon Chronos (zero-shot).

Measures. Prediction quality is the monthly rank IC: the correlation between forecast ranks and realised return ranks across stocks. Trading performance uses a monthly long-only portfolio of the top 5% of forecasts, equally weighted and traded at the close, with 57 bp round-trip costs for delivery trades. Returns are compared against the equal-weighted average of all filtered stocks.

4. Results

StockGPT predicts Indian returns out of sample, and its signal held up in 2019–2025 while momentum's weakened.

Selection period (2011–2018). At a one-day horizon, base StockGPT was the strongest predictor, with a rank IC of 0.071 against 0.033 for the other models. At the monthly horizon used for trading, it trailed plain 12-1 momentum (IC 0.031 against 0.079), but it cleared the reversal floor and was carried into the final test as the main model.

Final test (2019–2025), model frozen at 2010.

Signal	2011–2018 rank IC	2019–2025 rank IC
Base StockGPT	0.031	0.036 (t = 3.9)
12-1 momentum	0.079	0.040 (t = 2.5)

Momentum's IC halved out of sample; StockGPT's rose slightly and was steadier, positive in 65% of months.

- **Long-only top 5%, after costs.** 31.8% a year against 19.8% for the average filtered stock, an excess of +12.0% (t = 1.41, net Sharpe 0.73). Momentum's excess was +3.4%; LightGBM and Chronos were near zero.
- **Long-short, top minus bottom 10%.** 34.1% a year with a Sharpe of 2.11, against 23.6% (Sharpe 1.06) for momentum. Most of the short side cannot be traded in practice (see section 5).
- **Spanning test.** After controlling for momentum and reversal, StockGPT's long-short still earns +40.7% a year (t = 7.2). Its momentum loading is negative (−0.31), so the signal is distinct from momentum rather than a disguised version of it.

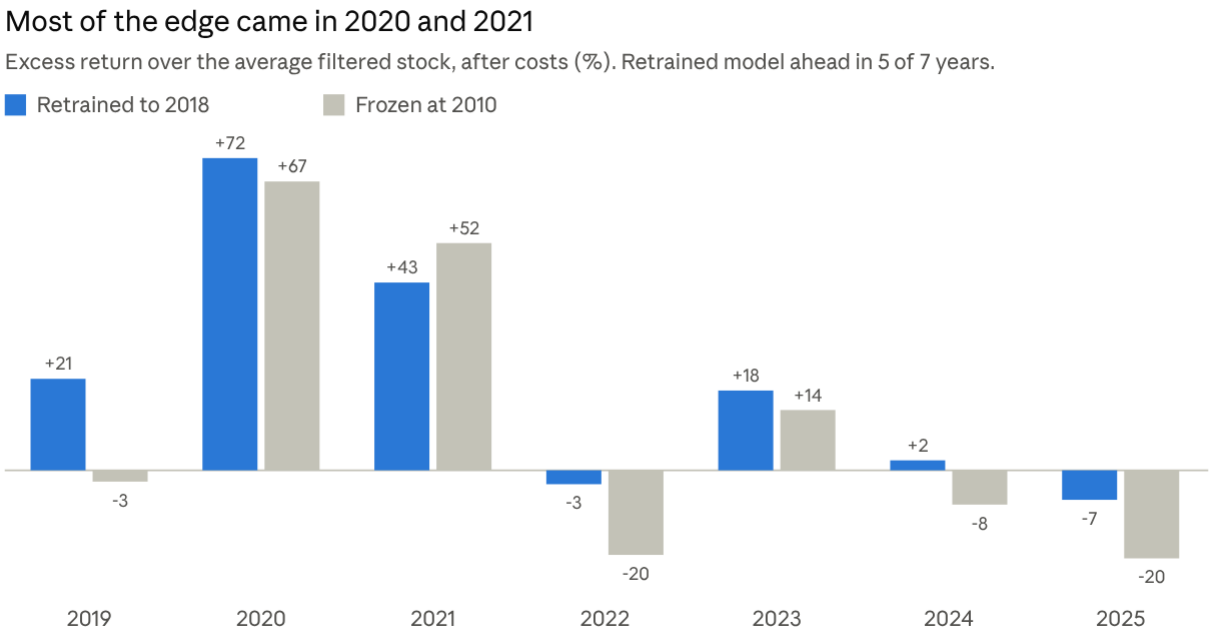
5. Robustness

Retraining on more recent data strengthens the signal, but the gains sit in smaller stocks and in two years.

Retrained model. Mai (2024) recommends retraining as new data arrives. As a robustness check, decided after the frozen results were seen, the model was retrained on 1995–2018 and run once on 2019–2025.

Long-only top 5%, after costs	Frozen at 2010	Retrained to 2018
Annual return	31.8%	**41.1%**
Excess over average stock	+12.0% (t = 1.41)	**+21.3% (t = 3.44)**
Net Sharpe ratio	0.73	**1.15**
Rank IC	0.036 (t = 3.9)	**0.067 (t = 6.0)**
Months with positive IC	65%	**80%**
Monthly turnover	55%	58%

Year by year.



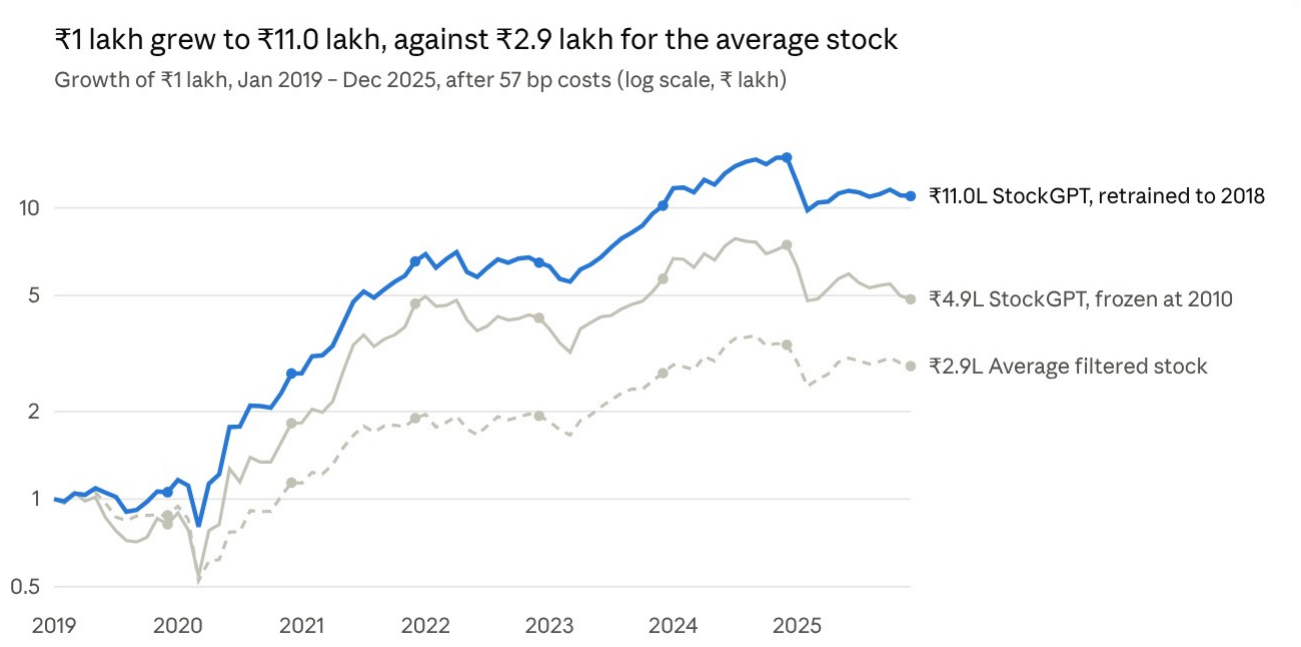
Final-test results, StockGPT NSE replication, 2019–2025, long-only top 5%, after 57 bp costs

2020 and 2021, the post-COVID small-cap rally, contributed about +115 percentage points of the retrained model's excess. Across the other five years its excess averaged roughly +6% a year, with 2022 and 2025 negative.

Large, liquid stocks only. Restricting the test to the roughly 180 F&O stocks a month (about 18 names a side, 13 bp futures costs), base StockGPT's long-short returns −2.1% a year after costs (t = −0.33). No signal tested was significant in this universe; momentum's +7.1% a year came with a 76.5% drawdown. The whole-universe result therefore comes from smaller, less liquid stocks, consistent with Mai's finding that returns fall sharply under value weighting.

6. Full backtest report

The retrained model turned ₹1 lakh into ₹11.0 lakh over 2019–2025, against ₹2.9 lakh for the average stock, with a smaller worst fall.



Growth of ₹1 lakh, StockGPT NSE replication, Jan 2019 – Dec 2025, long-only top 5%, after 57 bp costs

The curve also shows the main risk: after peaking in December 2024, the portfolio fell 34% by February 2025 and had not regained its peak by December 2025.

Portfolio statistics (monthly, after 57 bp costs; months dated by when the return was earned; Sharpe and Sortino without a risk-free rate)

Measure	Retrained to 2018	Frozen at 2010	Average filtered stock
Compounded annual return	**41.4%**	25.7%	16.4%
Growth of ₹1 lakh	**₹11.0 lakh**	₹4.9 lakh	₹2.9 lakh
Annual volatility	35.6%	43.6%	29.1%
Sharpe ratio	**1.15**	0.73	0.68
Sortino ratio	**2.30**	1.43	1.01
Maximum drawdown	**−34.2%** (Dec 2024 – Feb 2025, not recovered by Dec 2025)	−48.5% (Mar 2019 – Mar 2020, recovered Jun 2020)	−50.8% (Mar 2019 – Mar 2020, recovered Dec 2020)
Calmar ratio (return ÷ max drawdown)	**1.21**	0.53	0.32
Longest time below a previous peak	14 months	21 months	20 months

Measure	Retrained to 2018	Frozen at 2010	Average filtered stock
Best month	+45.3%	+55.2%	+24.3%
Worst month	−27.8%	−30.8%	−38.6%
Positive months	66%	59%	60%
Months beating the average stock	61%	48%	—

Trade statistics (each stock held for one month counts as one pick; before costs)

Measure	Retrained to 2018	Frozen at 2010
Stocks held each month	about 69	about 69
Total picks, 83 months	5,756	5,756
Winning picks	**54.1%**	50.8%
Average winning pick	+16.3%	+17.6%
Average losing pick	−11.4%	−12.5%
Profit factor (total gains ÷ total losses)	**1.69**	1.45
Monthly turnover	57%	54%

The edge comes from a modest hit rate (54% of picks win) combined with winners that are larger than losers, not from a high win rate. The 41.4% compounded return is close to the 41.1% average-month figure quoted earlier, which annualises the mean monthly return.

7. Limitations

- **Concentration in time.** Two strong years drive most of the excess return, and 2019–2025 was an unusually strong period for Indian small caps.
- **Smaller stocks and real costs.** The edge lives outside the F&O universe. Portfolios are equal-weighted, and the 57 bp cost does not capture market impact in thinly traded names, so results at real size would be lower.
- **High turnover.** About 55–58% of the portfolio changes each month, which raises the cost drag.
- **Shorting.** Outside F&O stocks the short side cannot be traded, so the long-short figures are descriptive only.
- **One run.** Each model was trained once with one seed and tested once; results were not re-run to improve them. The retrained model was added after the frozen results were seen and is reported as a robustness check.
- **Recent drawdown.** The worst fall, 34% from December 2024 to February 2025, had not been recovered by the end of 2025.

8. Conclusion

A language model that reads only past daily returns carries a real, out-of-sample signal in Indian equities. Its prediction quality held up in 2019–2025 while momentum's halved, its signal is distinct from momentum, and retraining on more recent data made it stronger (rank IC 0.067, t = 6.0). Turned into a long-only portfolio, the retrained model earned 41.1% a year after costs against 19.8% for the average stock. That edge is concentrated in smaller stocks and in 2020–2021, and its worst fall (34%) came late in the test, so the next steps are to test liquidity tiers and value weighting, and track the signal live before trading it at size.

References

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